Dynamic Competition for Sleepy Deposits

Mark L. Egan, Ali Hortaçsu, Nathan A. Kaplan, Adi Sunderam, Vincent Yao

Lucas Schmitz

February 18, 2026

Yale Economics

Outline

Motivation and Questions

Data and Stylized Facts

Model

Estimation Strategy

Counterfactual: eliminating sleepiness

Conclusion

Motivation

Retail depositors rarely shop for better terms.

- Stated reason for account closures
  - Less than 20% due to better rates
  - Most closures are idiosyncratic (e.g., inactivity, no longer needed, moving, death)
- Account turnover is low: new accounts are less than 15% of existing accounts per year

Possible implications of sleepiness:

- Competition: creates invest and harvest incentives that can increase/decrease competition
- Bank value: has impact on franchise value
- Financial stability: affects the impact of interest rate changes on franchise value

This paper

Question: How does depositor sleepiness shape deposit market competition, markups, and bank franchise value?

What they do

1. Estimate sleepiness ϕ_{kt}
2. Estimate depositor demand (Berry 1994) among active depositors
3. Recover bank costs via BBL

What they find

- ≈94% of depositors inactive each year
- Sleepiness raises average markups by **53%** (68 vs. 32 bps)
- Markups are **procyclical** — invest-harvest incentives vary with rates
- Sleepiness explains ≈**54%** of deposit franchise value

Related literature

- **Banking + IO:** Dick (2008), Egan et al. (2017), Xiao (2020), Wang et al. (2022)
  - Incorporates dynamics into supply and demand for bank deposits
- **Dynamic competition + Inertia in demand:** Beggs and Klemperer (1992), Dubé et al. (2009), MacKay and Remer (2024), Brown et al. (2023) and Einav et al. (2025)
  - Applies a model with sleepiness to the deposit market
- **Bank deposits:** Berger and Hannan (1991), Argyle et al. (2025), Drechsler et al. (2023)
  - Studies the impact of sleepiness in the deposit market

Outline

Motivation and Questions

Data and Stylized Facts

Model

Estimation Strategy

Counterfactual: eliminating sleepiness

Conclusion

Data sources

- Account-level turnover microdata: includes account closures/opening the reason and demographics
- FDIC Summary of Deposits (SoD): deposits at the bank$\times$county$\times$year level
- RateWatch: bank$\times$county$\times$year average rates by account type
- Call Reports: bank-level cost shifters (salary expenses, premises/fixed-asset expenses)
- Final estimation panel: 341,395 bank$\times$county$\times$year observations, 2002-2023

Deposit rates and spreads

Table 1: Summary Statistics

Variable	Observations	Mean	Median	SD
Bank-County Level:				
Deposit Rate (bps)	341,395	34	15	50
Deposit Spread (bps)	341,395	97	17	145
Deposit Market Share	341,395	0.16	0.10	0.18
County Deposits ($M)	341,395	405	84	4046
Bank Level:				
Wages ($1000s)	95,693	69	64	24
$100 \times \frac{\text{Fixed Expenses}}{\text{Total Assets}}$	95,693	0.39	0.36	0.18
Total Deposits ($M)	95,693	1874	171	30712

- Sample: $10k money market accounts
- Deposit spread ("price" of deposits): $\rho_{jkt} := R_t^F - r_{jkt}$

Fact 1: low turnover

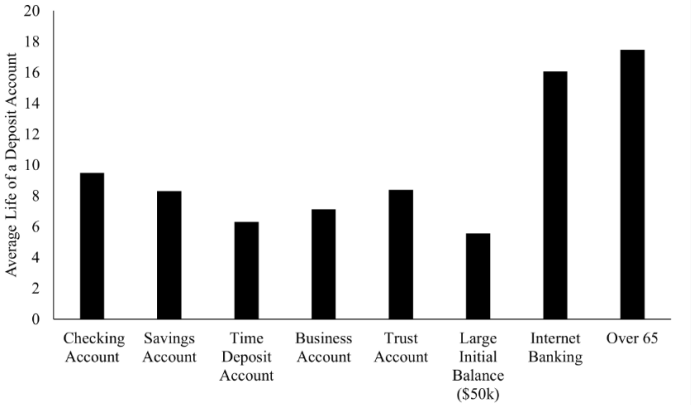


Figure 1: Avg. Life $= \frac{1}{\text{Turnover Rate}}$ where turnover rate is the fraction of newly opened accounts in a year.

- Cross-sectional patterns suggest frictions (search/switching/inattention), not just persistent tastes

Fact 2: reasons for account closures

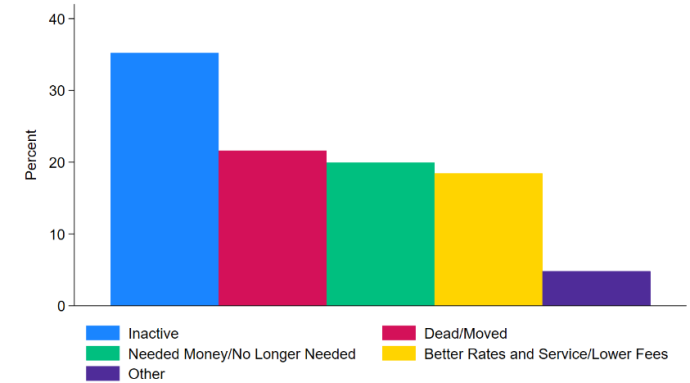


Figure 2

- Banks record closure reasons for about 75% of closures
- Most reasons are idiosyncratic

Outline

Motivation and Questions

Data and Stylized Facts

Model

Estimation Strategy

Counterfactual: eliminating sleepiness

Conclusion

Model overview

Dynamic deposit market with **active vs. inactive** depositors.

- Each period depositors are either awake (active) or asleep (inactive).
- Active depositors choose a bank via discrete choice; inactive depositors keep last period's bank.
- Banks set deposit rates/spreads dynamically to maximize franchise value.
- Sleepiness generates **invest vs. harvest** incentives:
  - Harvest: inertia lowers elasticity $\Rightarrow$ higher spreads
  - Invest: capturing a depositor today raises future deposits

Depositor utility and active demand

Active depositor i choosing bank j in market k at time t:

$$u_{ijkt} = \alpha\rho_{jkt} + \delta_{jkt} + \varepsilon_{ijkt}, \qquad \rho_{jkt} := R_t^F - r_{jkt}, \quad \alpha < 0.$$

With Type-I extreme value ε_{ijkt}, active market share:

$$s_{jkt}^{Active} = \frac{\exp(\alpha\rho_{jkt} + \delta_{jkt})}{\sum_{\ell \in J_k} \exp(\alpha\rho_{\ell kt} + \delta_{\ell kt})}. \tag{3}$$

Sleepiness (inactivity) process

- Depositor activity status: $D_{it} = 1$ if active, 0 if inactive.
- Activity given by:

$$D_{it}^* = S'_{k(i)t}\Gamma + X'_{it}\Theta + \eta_{it}, \qquad D_{it} = 1(D_{it}^* > 0). \tag{2}$$

- S_{kt}: lagged federal funds rate R_{t-1}^F; $X_{\ell kt}$: depositor-type indicators - age > 65, initial balance $<$ \$50,000, internet banking enrollment
- Market-level inactivity share:

$$\phi_{kt} = 1 - \mathbb{E}[D_{it} \mid S_{kt}].$$

- Interpretation: reduced-form inattention / switching frictions; empirically ϕ_{kt} varies with rates and demographics.

Total deposits: active + inactive components

Total deposits held by bank j in market k at time t:

$$Dep_{jkt} = \underbrace{(1-\phi_{kt})M_{kt}s_{jkt}^{Active}}_{\text{Active depositor demand}} + \underbrace{\phi_{kt}(1+r_{jk,t-1})Dep_{jk,t-1}}_{\text{Inactive depositor demand}}. \tag{4}$$

- Active flow: $(1-\phi_{kt})M_{kt}s_{jkt}^{Active}$.
- Inactive stock: last period deposits roll over and earn interest.

Banks: flow profits and franchise value

Banks invest deposits at bank-level return R_{jt} and pay deposit rate r_{jkt} and marginal servicing cost c_{jkt}.

$$\pi_{jt} = \sum_{k \in K} Dep_{jkt} \Big((R_{jt} - R_t^F) + \rho_{jkt} - c_{jkt} \Big)$$

Franchise value:

$$V_{jt} = \mathbb{E}\Big[\sum_{s=t}^{\infty} \beta^{s-t} \pi_{js} \Big].$$

Equilibrium: stationary Markov perfect equilibrium in pure strategies (BBL approach).

Equilibrium: dynamic rate-setting game (BBL)

State S_{kt}: exogenous public (Ω_{kt}: fed funds rate, δ_{jkt}), endogenous public (W_{kt}: market shares), private (χ_{kt}: iid cost shocks).

Bank j's strategy: $\sigma_j : \mathsf{S}_{kt} \times \chi_{jkt} \to \rho_{jkt}$. Franchise value satisfies:

$$V(\mathsf{S}_{kt}, \boldsymbol{\sigma}) = \mathbb{E}_{\chi_{jt}, R_{jt}} \Big[Dep(\mathsf{S}_{kt}, \boldsymbol{\sigma}) \big(R_{jt} - R_t^F + \sigma_j(\mathsf{S}_{kt}, \chi_{jt}) - c(\mathsf{S}_{kt}, \chi_{jt}) \big) \,\Big|\, \mathsf{S}_{kt} \Big]$$

$$+ \beta \, \mathbb{E}_{\mathsf{S}_{kt+1}, \boldsymbol{\chi}_{t+1}} \Big[V(\mathsf{S}_{kt+1}, \boldsymbol{\sigma}) \,\Big|\, \boldsymbol{\sigma}, \mathsf{S}_{kt} \Big]. \tag{5}$$

Equilibrium: each bank prefers its strategy to any alternative Markov strategy,

$$V(\mathsf{S}, \boldsymbol{\sigma}) \geq V(\mathsf{S}, \sigma_j', \boldsymbol{\sigma}_{-j}) \quad \forall\, \mathsf{S},\ \forall\, \sigma_j',\ \forall\, j.$$

Dynamic vs. static pricing: invest–harvest decomposition

Differentiating the Bellman (simplified: constant R_t^F, constant qualities) yields:

$$\underbrace{\rho_j - c}_{\text{markup}} = \underbrace{\frac{-1}{\alpha(1 - s_j^{Active})}}_{\text{static pricing}} + \underbrace{\frac{\phi}{1-\phi}\frac{s_{j,-}}{s_j^{Active}}\frac{-1}{\alpha(1 - s_j^{Active})}}_{\text{harvesting incentive}} + \underbrace{\beta\,\nabla V_j(\mathrm{s} \mid \boldsymbol{\delta}, \mathrm{c}) \cdot \mathrm{d}_j}_{\text{investing incentive}}, \tag{6}$$

where d_j is the vector of diversion ratios from bank j.

- **Harvesting (↑ markups):** captive sleepy base lowers residual demand elasticity; strength increasing in ϕ and lagged share $s_{j,-}$.
- **Investing (↓ markups):** winning a depositor today yields future rents
- Two channels push in **opposite directions** ⇒ net effect is an empirical question.

Outline

Motivation and Questions

Data and Stylized Facts

Model

Estimation Strategy

Counterfactual: eliminating sleepiness

Conclusion

Estimation: three steps

1. **Sleepiness** ϕ_{kt}, two methods:
   - Use account openings/turnover microdata
   - Use deposit autocorrelation in aggregate deposits (SoD)
2. **Demand** (active depositors): compute active market shares using ϕ_{kt}, estimate (α, δ_{jkt}) via Berry (1994).
3. **Costs / supply** (dynamic game): estimate reduced-form spread policy functions; recover cost parameters via BBL using that observed policies maximize franchise value.

Step 1: identifying sleepiness from account openings

Under the model, new accounts identify active demand:

$$NewDep_{\ell jkt} = (1 - \phi(S_{kt}, X_{\ell kt}))M_{\ell kt} s_{jkt}^{Active}(1 - s_{jk,t-1}).$$

Rearrangement yields a moment linking turnover to sleepiness:

$$1 - \frac{NewDep_{\ell jkt}}{Dep_{\ell jkt}} \frac{1}{1 - s_{jk,t-1}} = \phi(S_{kt}, X_{\ell kt}) \frac{Dep_{\ell jk,t-1}}{Dep_{\ell jkt}} + e_{\ell jkt}. \tag{8}$$

Key finding: $\approx 94\%$ of depositors are inactive each year; sleepiness falls when the lagged fed funds rate is higher (people "wake up" when returns to shopping rise).

Sleepiness vs. persistence

Step 1: autocorrelation and control function

Autocorrelation in deposits confounds sleepiness with persistent quality.

- Control function: regress spreads on cost shifters (salaries, fixed expenses) to recover latent demand:

$$\rho_{jkt} = \boldsymbol{\lambda}' \mathrm{Z}_{jt} + v_{jkt}.$$

- Second stage: regress deposits on lagged deposits and the control:

$$Dep_{jkt} = (\boldsymbol{\Upsilon}_1' \mathrm{S}_{kt} + \boldsymbol{\Upsilon}_2' \mathrm{X}_{kt})(1 + R^F_{t-1} - \rho_{jk,t-1}) Dep_{jk,t-1} + H(v_{jkt}) + \iota_{jkt}.$$

Identifying assumptions

Table 3: Sleepiness Estimates

	Microdata only		SoD only		Combined	
	(1)	(2)	(3)	(4)	(5)	(6)
$\Upsilon_{1,\text{cons}}$	0.868***	0.877***	0.977***	0.979***	0.904***	0.945***
	(0.008)	(0.010)	(0.001)	(0.002)	(0.007)	(0.006)
Υ_{1,R_t^F}		-0.416*		-0.105***		-0.355**
		(0.249)		(0.041)		(0.169)
$\Upsilon_{2,\text{over 65}}$		0.0526***				0.017***
		(0.005)				(0.005)
$\Upsilon_{2,\text{over \$50k}}$		-0.094***				-0.130***
		(0.006)				(0.007)
$\Upsilon_{2,\text{internet banking}}$		0.036***				0.002
		(0.007)				(0.007)
Observations	8,878	8,878	341,395	341,395	350,273	350,273
Within R-squared	0.966	0.966	0.963	0.963	0.962	0.966
Bank FE			X	X	X	X
County-Year FE			X	X	X	X

Step 2: demand among active depositors

- Compute active deposits using $\hat{\phi}_{kt}$:

$$Dep_{jkt}^{Active} := \max\left\{0,\ Dep_{jkt} - \hat{\boldsymbol{\Upsilon}}'\begin{pmatrix} S_{kt} \\ \tilde{X}_{kt} \end{pmatrix}(1 + R_{t-1}^{F} - \rho_{jk,t-1})Dep_{jk,t-1}\right\}.$$

- Use Berry (1994) to estimate (α, δ) from:

$$\log(s_{jkt}^{Active}) = \alpha\rho_{jkt} + \delta_j + \mu_{kt} + e_{jkt}. \tag{11}$$

- Endogeneity of ρ_{jkt}: use cost shifters Z_{jt} as instruments

Table 4: Demand Estimates

	Total Market (No Inertia)		Active Market	
	OLS	CF	OLS	CF
	(1)	(2)	(3)	(4)
ρ_{jkt}	-3.17***	-290.4***	-16.2***	-388.7***
	(1.14)	(15.3)	(2.05)	(29.3)
Observations	294,809	294,809	294,809	294,809
Rate Elasticity of Demand	0.004	0.392	0.022	0.525
R^2	0.733	0.734	0.625	0.626
Bank fixed effects	✓	✓	✓	✓
County-Year fixed effects	✓	✓	✓	✓

Step 3a: costs, state space, and policy function (BBL)

- Parameterize net marginal costs $(R_{jt} - c(S_{kt}, \chi_{jkt})$ as:

$$c_{jkt} := \omega + \zeta R_t^F + \gamma' \mathsf{Z}_{jt} + \chi_{jkt}.$$

  R_t^F included because lending returns may vary with the short rate; also isolates markup variation driven by invest-versus-harvest incentives.

- **State space** S_{kt}: (i) R_t^F, (ii) R_{t-1}^F, (iii) lagged market shares, (iv) cost shifters Z_{kt}, (v) product qualities $\boldsymbol{\delta}_{kt}$, (vi) county demographics, (vii) private cost shocks χ_{jkt}.

Step 3b: state transitions and forward simulation (BBL)

- Recover policy functions
  - $\sigma : S_{kt} \times \chi_{jkt} \rightarrow \rho_{jkt}$: first-order polynomial in all states + sums of competitors' characteristics (costs, lagged shares)
- State transitions:
  - Endogenous states: deterministic transition
  - δ_{jkt}, Z_{jt} remain constant
  - Short term rate from FED's estimates
- Forward simulation:
  - Given policies and state transitions, forward-simulate to obtain value functions.
  - Pick cost parameters that rationalize observed policies
  - Perturb policies and minimize the expected sum of squared violations

Table 6: Marginal Cost Estimates

ζ	0.695
	(0.030)
ω	-0.008
	(0.003)
γ_1	5.139×10^{-6}
	(1.190×10^{-6})
γ_2	0.013
	(0.325)

Outline

Motivation and Questions

Data and Stylized Facts

Model

Estimation Strategy

Counterfactual: eliminating sleepiness

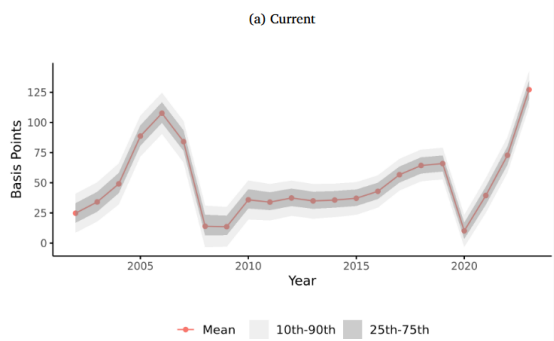
Conclusion

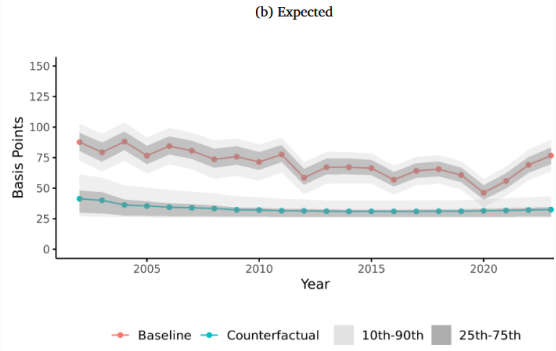
Counterfactual: static competition ($\phi = 0$)

Compare the estimated dynamic model against a counterfactual in which **all depositors are always active** ($\phi = 0$).

- Competition becomes static Bertrand-Nash: no invest-versus-harvest tradeoff.
- Isomorphic to the limiting case of the UK FCA policy requiring full pass-through of short rates into deposit rates.
- Intuition: eliminating sleepiness removes the stock of captive future depositors $\Rightarrow$ no intertemporal return to attracting depositors today $\Rightarrow$ competition collapses to static Bertrand.

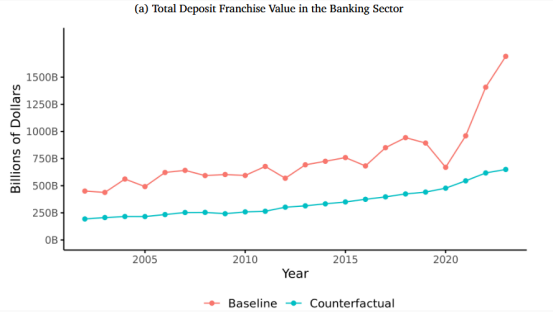
Counterfactual: markups and concentration

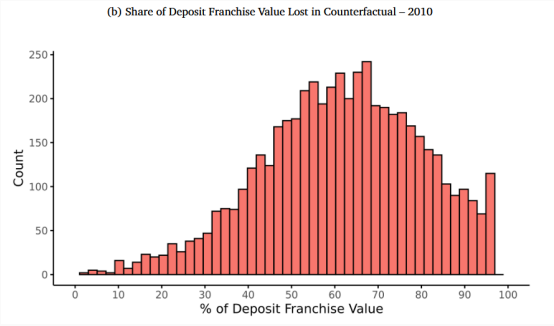




- Dynamic model: average markup $\approx$ 68 bps; static counterfactual: $\approx$ 32 bps (**53% lower**).
- Dynamic competition eliminates the positive concentration–markup (HHI) relationship: banks harvest sleepy bases even in low-HHI markets.

Counterfactual: franchise value decomposition





- Franchise value decreases by $\approx$ **54%** ($\approx$ \$1 trillion if implemented in 2023).
- Heterogeneity: low-quality and high-cost banks lose the most from eliminating sleepiness.

  Heterogeneity
- Financial stability: for JPM, BofA, Wells Fargo, default probabilities rise $\approx$10 pp in normal times and $>$**20 pp** during the 2022–23 tightening cycle.

Outline

Motivation and Questions

Data and Stylized Facts

Model

Estimation Strategy

Counterfactual: eliminating sleepiness

Conclusion

Conclusion

- Most retail depositors are sleepy
- Sleepiness raises markups and makes them procyclical
- Sleepiness explains a large share of deposit franchise value and supports stability.
- Policies aimed at improving deposit rate pass-through face a fundamental tradeoff: lower consumer spreads vs. reduced bank resilience.

Comments

- There are persistent idiosyncratic tastes
- Banks are multi-product firms; low prices might not only reflect invest-harvest but also cross-subsidization from other products (e.g., credit cards, mortgages)
- Substitution patterns: not heterogeneity in elasticities nor sleepiness
- Banks use introductory pricing, the invest-harvest tradeoff might not be the most important
  - Price discrimination could be a first order concern
  - Spread increasing on time
- Intensive margin: people do not close accounts, they might stop using them

Sleepiness vs. persistent preferences

Figure 3: Sleepiness versus Unobserved Preference Heterogeneity: Evidence from Market Entry

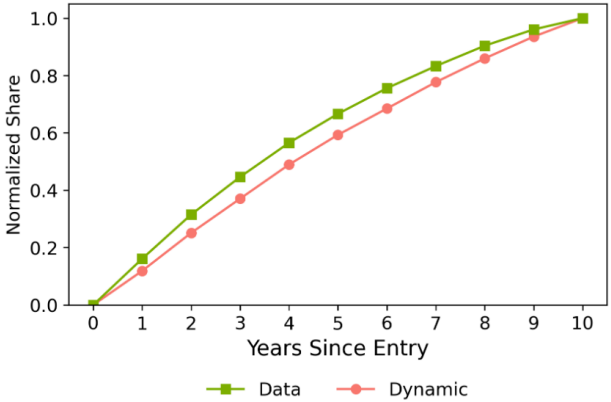


Figure 5

Control function: identifying assumptions

Problem: naive regression of deposits on lagged deposits is biased—banks with persistently high quality δ_{jkt} have both high deposits and high spreads, confounding sleepiness with persistent demand.

Control function approach (eqs. 9–10):

$$\text{First stage:} \quad \rho_{jkt} = \lambda Z_{jt} + \underbrace{v_{jkt}}_{\text{latent demand}} \tag{9}$$

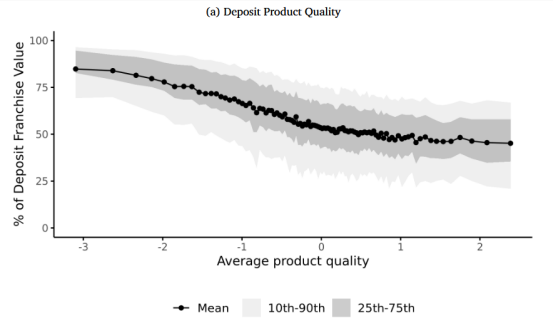
$$\text{Second stage:} \quad Dep_{jkt} = (\boldsymbol{\Upsilon}_1' S_{kt} + \boldsymbol{\Upsilon}_2' X_{kt})(1 + R^F_{t-1} - \rho_{jk,t-1}) Dep_{jk,t-1} + H(v_{jkt}) + \iota_{jkt} \tag{10}$$

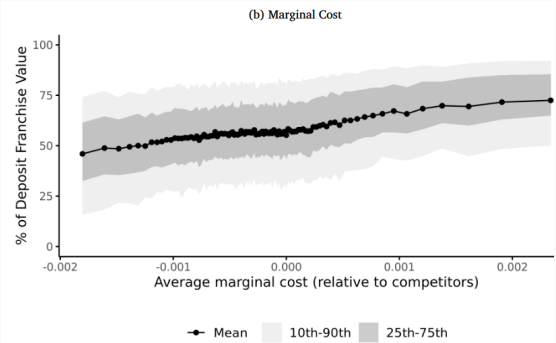
Two identifying assumptions:

1. Control function correctly specified: $\mathbb{E}[v_{jkt} \mid Z_{jt}] = 0$. Cost shifters Z_{jt} (salaries, fixed expenses) affect spreads but not demand directly.
2. Latent demand correctly specified: $\mathbb{E}[(1 + R^F_{t-1} - \rho_{jk,t-1}) Dep_{jk,t-1} \cdot \iota_{jkt}] = 0$. After controlling for $H(v_{jkt})$, lagged deposits are uncorrelated with the residual.

Back

Heterogeneity: who benefits most from sleepiness?





Back